Homework 3 Pseudocode

Bayou City Family Health Patient Registration Form

- Name: Brian Nwabuoku
- Class: MIS 7375 Transaction Processing Systems I
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Main Program

MAIN

DO BEGINNING
DO THE MIDDLE
DO THE END
END MAIN

DO BEGINNING

DISPLAY THE HTML FORM (homework3.html)
SHOW TODAYS DATE IN THE BANNER
LOAD THE STATE LIST FROM homework3-states.js INTO THE STATE BOX
    PUT "Select" FIRST WITH NO VALUE SO THE USER HAS TO PICK ONE
SHOW THE SLIDER VALUE NEXT TO THE BAR
SET FORM PASSED TO NO
HIDE THE SUBMIT BUTTON AND TURN IT OFF
DISPLAY "CLICK REVIEW AND VALIDATE WHEN YOU'RE DONE" IN THE STATUS LINE
SET-CONDITIONS (oninput, onblur, onchange, onclick, onsubmit and onreset)
END BEGINNING

DO MIDDLE (javascript, homework3.js)

IF ANYTHING ON THE FORM CHANGED
    CALL FORM-CHANGED FOR THE FIELD THAT CHANGED
ENDIF
IF THE USER IS TYPING IN THE SSN BOX
    CALL FORMAT-SSN
ENDIF
IF THE USER IS TYPING IN THE USER ID BOX OR A PASSWORD BOX
    CALL CHECK-NOW FOR THAT FIELD
ENDIF

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IF THE USER ENTERED AND THEN LEFT A FIELD
        CALL SHOW-RESULT FOR THAT FIELD
ENDIF
IF THE USER PICKED A STATE OR A RADIO BUTTON
        CALL SHOW-RESULT FOR THAT FIELD
ENDIF
IF THE USER MOVED THE SLIDER
        CALL SHOW-SLIDER-VALUE
ENDIF
IF THE REVIEW AND VALIDATE BUTTON WAS CHOSEN
        CALL REVIEW-AND-VALIDATE
ENDIF
IF THE SUBMIT BUTTON WAS CHOSEN
        LET THE BROWSER CHECK REQUIRED FIELDS AND PATTERNS
        IF THE BROWSER CHECKS PASS
                CALL REVIEW-AND-VALIDATE ONE MORE TIME
                IF NO ERROR CONDITIONS
                        DISPLAY THANK YOU PAGE
                ELSE
                        DO NOT SEND THE FORM
                ENDIF
        ENDIF
ENDIF
IF THE RESET BUTTON WAS CHOSEN
        CLEAR THE FORM
        CLEAR THE REVIEW AREA AND EVERY ERROR MESSAGE
        CALL LOCK-FORM
        PUT THE STATUS LINE BACK TO "CLICK REVIEW AND VALIDATE WHEN YOU'RE
DONE"
        PUT THE SLIDER VALUE BACK TO 5
ENDIF
END MIDDLE
```

THE-END

```
SHOW THE THANK YOU PAGE (thankyou.html)
DISPLAY "THANK YOU FOR YOUR SUBMISSION. WE WILL BE CONTACTING YOU
SHORTLY"
GIVE A LINK BACK TO THE FORM
END THE-END
```

Validate Button

REVIEW-AND-VALIDATE

```
CALL RUN-CHECKS AND KEEP THE RESULTS
CALL SHOW-REVIEW WITH THE RESULTS
SET ERROR COUNT TO 0
LOOP THROUGH EVERY RESULT UNTIL NONE LEFT
        IF THE RESULT IS AN ERROR MESSAGE
                ADD 1 TO ERROR COUNT
        ENDIF
END LOOP
IF ERROR COUNT IS 0
        SET FORM PASSED TO YES
        SHOW THE SUBMIT BUTTON AND TURN IT ON
        DISPLAY "EVERYTHING CHECKS OUT, CLICK SUBMIT" IN THE STATUS LINE
ELSE
        SET FORM PASSED TO NO
        HIDE THE SUBMIT BUTTON AND TURN IT OFF
        DISPLAY ERROR COUNT AND "STILL NEED FIXING" IN THE STATUS LINE
ENDIF
RETURN ERROR COUNT
END REVIEW-AND-VALIDATE
```

RUN-CHECKS

```
LOOP THROUGH EVERY TEXT BOX UNTIL NONE LEFT
        CALL TIDY FOR THAT BOX
END LOOP
LOOP THROUGH EVERY FIELD THAT HAS A CHECK UNTIL NONE LEFT
        CALL THE CHECK FOR THAT FIELD
        KEEP WHAT IT RETURNS IN THE RESULTS
        PUT WHAT IT RETURNS IN THE ERROR SPOT UNDER THE FIELD
END LOOP
RETURN THE RESULTS
END RUN-CHECKS
```

LOCK-FORM

```
IF FORM PASSED IS YES
        DISPLAY "YOU CHANGED SOMETHING, CLICK REVIEW AND VALIDATE AGAIN" IN
THE STATUS LINE
ENDIF
```

SET FORM PASSED TO NO
HIDE THE SUBMIT BUTTON AND TURN IT OFF
END LOCK-FORM

FORM-CHANGED (field that changed)

CALL LOCK-FORM
IF THE FIELD HAS NO CHECK OR ALREADY CHECKS ITSELF AS THE USER TYPES
    DO NOTHING
ELSE IF AN ERROR MESSAGE IS SHOWING UNDER THE FIELD
    CALL CHECK-NOW FOR THAT FIELD
ENDIF
END FORM-CHANGED

Helper Modules

GET-FIELD

FIND THE FIELD IN THE FORM BY ITS NAME (NOT ITS ID)
RETURN THE FIELD
END GET-FIELD

GET-VALUE

CALL GET-FIELD
RETURN THE VALUE WITHOUT THE EXTRA SPACES ON THE ENDS
LEAVE THE BOX THE WAY THE USER TYPED IT
END GET-VALUE

TIDY (field name)

IF THE FIELD IS NOT A TEXT BOX
    DO NOTHING
ENDIF
TRIM THE EXTRA SPACES OFF THE ENDS
CHECK THE KIND OF FIELD
    CASE OF EMAIL OR USER ID
        CHANGE IT TO LOWERCASE
    END CASE
    CASE OF MIDDLE INITIAL
        CHANGE IT TO UPPERCASE
    END CASE
    CASE OF ZIP WITH 5 DIGITS DASH 4 DIGITS

```
            CUT IT TO THE FIRST 5 DIGITS
        END CASE
        CASE OF PHONE WITH 10 DIGITS
            TAKE OUT EVERYTHING THAT IS NOT A DIGIT
            CHANGE IT TO 000-000-0000
        END CASE
REDISPLAY THE CLEANED UP VALUE IN THE BOX
END TIDY
```

CHECK-NOW (field name)

```
CALL THE CHECK FOR THAT FIELD
PUT WHAT IT RETURNS IN THE ERROR SPOT UNDER THE FIELD (FOUND BY ID)
IF THIS IS PASSWORD 1 AND PASSWORD 2 IS NOT EMPTY
    CALL CHECK-PASSWORD-2 AND SHOW ITS RESULT TOO
ENDIF
END CHECK-NOW
```

SHOW-RESULT (field name)

```
CALL TIDY FOR THAT FIELD
CALL CHECK-NOW FOR THAT FIELD
END SHOW-RESULT
```

FORMAT-SSN (runs as the user types)

```
KEEP ONLY THE DIGITS THE USER TYPED UP TO 9
IF THERE ARE MORE THAN 3 DIGITS
    PUT A DASH AFTER THE THIRD DIGIT
ENDIF
IF THERE ARE MORE THAN 5 DIGITS
    PUT A DASH AFTER THE FIFTH DIGIT
ENDIF
REDISPLAY THE SSN WITH ITS DASHES IN THE BOX
IF THE USER TYPED SOMETHING THAT IS NOT A NUMBER
    DISPLAY "SSN can only have numbers" UNDER THE BOX
ELSE IF THERE ARE LESS THAN 9 DIGITS
    DISPLAY HOW MANY DIGITS ARE IN SO FAR UNDER THE BOX
ELSE
    CLEAR THE MESSAGE UNDER THE BOX
ENDIF
END FORMAT-SSN
```

SHOW-SLIDER-VALUE

DISPLAY THE SLIDER NUMBER AS "N out of 10" NEXT TO THE BAR
END SHOW-SLIDER-VALUE

SHOW-STATUS (message, color)

DISPLAY THE MESSAGE IN THE STATUS LINE UNDER THE BUTTONS
END SHOW-STATUS

Field Checks

EVERY CHECK

CHECK ONLY ITS OWN FIELD
RETURN THE ERROR MESSAGE IF SOMETHING IS WRONG
RETURN NOTHING IF THE FIELD IS GOOD
LEAVE THE ERROR SPOT TO CHECK-NOW AND THE BOX TO TIDY

CHECK-FIRST-NAME

IF FIRST NAME IS EMPTY
    RETURN "First name is required"
ELSE IF IT HAS ANYTHING BUT LETTERS APOSTROPHES AND DASHES OR IS OVER 30 CHARS
    RETURN "Letters, apostrophes and dashes only"
ENDIF
END CHECK-FIRST-NAME

CHECK-MIDDLE-INIT

IF MIDDLE INITIAL IS NOT EMPTY AND IS NOT ONE LETTER
    RETURN "Middle initial has to be one letter"
ENDIF
END CHECK-MIDDLE-INIT

CHECK-LAST-NAME

IF LAST NAME IS EMPTY
    RETURN "Last name is required"
ELSE IF IT HAS ANYTHING BUT LETTERS APOSTROPHES DASHES OR NUMBERS 2 TO 5
    RETURN "Letters, apostrophes, dashes and numbers 2 to 5 only"
ENDIF

```
END CHECK-LAST-NAME
```

CHECK-DOB

```
IF DATE OF BIRTH IS EMPTY
        RETURN "Date of birth is required"
ELSE IF IT IS NOT MM/DD/YYYY
        RETURN "Please use MM/DD/YYYY"
ELSE
        SPLIT IT INTO MONTH DAY AND YEAR
        IF THAT DAY DOES NOT EXIST (LIKE 02/30)
                RETURN "That date doesn't exist"
        ENDIF
        GET TODAYS DATE (SAME DATE SHOWN IN THE BANNER)
        SET OLDEST DATE TO TODAY MINUS 120 YEARS
        IF DATE OF BIRTH IS AFTER TODAY
                RETURN "Can't be in the future"
        ELSE IF DATE OF BIRTH IS BEFORE THE OLDEST DATE
                RETURN "Can't be more than 120 years ago"
        ENDIF
ENDIF
END CHECK-DOB
```

CHECK-SSN

```
TAKE THE DASHES OUT
IF SSN IS EMPTY
        RETURN "Social Security number is required"
ELSE IF IT HAS ANYTHING BUT NUMBERS
        RETURN "SSN can only have numbers"
ELSE IF IT IS NOT 9 DIGITS
        RETURN "SSN needs 9 digits"
ENDIF
END CHECK-SSN
```

CHECK-ADDRESS-1

```
IF ADDRESS 1 IS EMPTY
        RETURN "Address line 1 is required"
ELSE IF IT IS LESS THAN 2 OR MORE THAN 30 CHARS
        RETURN "Needs 2 to 30 characters"
ELSE IF IT HAS ANYTHING BUT LETTERS NUMBERS SPACES AND . , # ' - /
        RETURN "Letters, numbers, spaces and . , # ' - / only"
ENDIF
```

END CHECK-ADDRESS-1

CHECK-ADDRESS-2

IF ADDRESS 2 IS EMPTY
    RETURN NOTHING (IT IS NOT REQUIRED)
ELSE IF IT IS LESS THAN 2 OR MORE THAN 30 CHARS
    RETURN "Needs 2 to 30 characters or leave it blank"
ELSE IF IT HAS ANYTHING BUT LETTERS NUMBERS SPACES AND . , # ' - /
    RETURN "Letters, numbers, spaces and . , # ' - / only"
ENDIF
END CHECK-ADDRESS-2

CHECK-CITY

IF CITY IS EMPTY
    RETURN "City is required"
ELSE IF IT IS LESS THAN 2 OR MORE THAN 30 CHARS
    RETURN "Needs 2 to 30 characters"
ELSE IF IT HAS ANYTHING BUT LETTERS SPACES PERIODS APOSTROPHES AND DASHES
    RETURN "Letters, spaces, periods, apostrophes and dashes only"
ENDIF
END CHECK-CITY

CHECK-STATE

IF STATE IS STILL "Select"
    RETURN "Please pick a state"
ENDIF
END CHECK-STATE

CHECK-ZIP

IF ZIP IS EMPTY
    RETURN "Zip code is required"
ELSE IF IT IS NOT 5 DIGITS OR 5 DIGITS DASH 4 DIGITS
    RETURN "Zip code needs 5 digits"
ENDIF
END CHECK-ZIP

CHECK-PHONE

IF PHONE IS EMPTY

```
        RETURN "Phone number is required"
ELSE IF IT IS NOT 10 DIGITS
        RETURN "Phone number needs 10 digits"
ENDIF
END CHECK-PHONE
```

CHECK-EMAIL

```
IF EMAIL IS EMPTY
        RETURN "Email address is required"
ELSE IF IT IS NOT name@domain.tld
        RETURN "Email needs to look like name@domain.com"
ENDIF
END CHECK-EMAIL
```

CHECK-USER-ID (runs as the user types)

```
IF USER ID IS EMPTY
        RETURN "User ID is required"
ELSE IF IT HAS A SPACE
        RETURN "No spaces"
ELSE IF IT STARTS WITH A NUMBER
        RETURN "Can't start with a number"
ELSE IF IT HAS ANYTHING BUT LETTERS NUMBERS UNDERSCORES AND DASHES
        RETURN "Letters, numbers, underscores and dashes only"
ELSE IF IT DOES NOT START WITH A LETTER
        RETURN "Has to start with a letter"
ELSE IF IT IS LESS THAN 5 OR MORE THAN 20 CHARS
        RETURN "Needs 5 to 20 characters"
ENDIF
END CHECK-USER-ID
```

CHECK-PASSWORD-1 (runs as the user types)

```
IF PASSWORD IS EMPTY
        RETURN "Password is required"
ENDIF
START A LIST OF WHAT IS MISSING
IF LESS THAN 8 OR MORE THAN 30 CHARS ADD "8 to 30 characters"
IF NO UPPERCASE LETTER ADD "an uppercase letter"
IF NO LOWERCASE LETTER ADD "a lowercase letter"
IF NO NUMBER ADD "a number"
IF NO SPECIAL CHARACTER ADD "a special character"
IF IT HAS A DOUBLE QUOTE
```

```
        RETURN "No double quotes"
ELSE IF THE LIST IS NOT EMPTY
        RETURN "Still needs" AND THE LIST
ELSE IF IT HAS THE USER ID IN IT
        RETURN "Can't include your user ID"
ELSE IF IT HAS THE FIRST OR LAST NAME IN IT
        RETURN "Can't include your name"
ENDIF
END CHECK-PASSWORD-1
```

CHECK-PASSWORD-2 (runs as the user types)

```
IF PASSWORD 2 IS EMPTY
        RETURN "Please type your password again"
ELSE IF PASSWORD 2 DOES NOT EQUAL PASSWORD 1
        RETURN "Passwords do not match"
ENDIF
END CHECK-PASSWORD-2
```

CHECK-RADIO (gender, vaccinations, insurance)

```
IF NO BUTTON IN THE GROUP IS CHOSEN
        RETURN "Please answer this question"
ENDIF
END CHECK-RADIO
```

CHECK-SYMPTOMS

```
IF THE TEXT AREA HAS A DOUBLE QUOTE
        RETURN "Please leave out double quotes"
ENDIF
END CHECK-SYMPTOMS
```

Review

SHOW-REVIEW (results)

```
DISPLAY HEADING "PLEASE REVIEW YOUR INFORMATION"
LOOP THROUGH EVERY LINE OF THE REVIEW UNTIL NONE LEFT
        DISPLAY THE FIELD LABEL FROM THE PAGE (NOT THE VARIABLE NAME)
        CHECK THE KIND OF DATA
                CASE OF NAME
                        DISPLAY FIRST MI AND LAST TOGETHER
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```
                END CASE
                CASE OF ADDRESS
                        DISPLAY ADDRESS 1, ADDRESS 2 IF THERE IS ONE, CITY STATE
AND ZIP
                END CASE
                CASE OF SSN
                        IF THE SSN CHECK PASSED
                                DISPLAY ONLY THE LAST 4 DIGITS
                        ELSE
                                DISPLAY "(hidden)"
                        ENDIF
                END CASE
                CASE OF A RADIO BUTTON
                        DISPLAY THE WORDING OF THE BUTTON THAT WAS CHOSEN
                END CASE
                CASE OF A CHECKBOX
                        IF THIS ITEM WAS CHECKED
                                DISPLAY "Y"
                        ELSE
                                DISPLAY "N"
                        ENDIF
                END CASE
                CASE OF THE SLIDER
                        DISPLAY "N out of 10"
                END CASE
                CASE OF THE PASSWORD
                        DISPLAY STARS ONLY
                END CASE
                CASE OF BUTTONS
                        DO NOTHING
                END CASE
                DEFAULT CASE
                        DISPLAY VALUE
                END CASE
        IF THE CHECK FOR THIS LINE PASSED
                DISPLAY "pass"
        ELSE
                DISPLAY "ERROR" AND THE MESSAGE
        ENDIF
END LOOP
END SHOW-REVIEW
```